

Submission: Upload your work to GitHub repo till the next practice lesson

Online Shop (Continued)

Building REST APIs with DRF

Upgrade your `shop-back` project from Lab 8 by integrating Django REST Framework.

Tasks (5)

1. Install Django REST Framework

- Install `djangorestframework` using `pip`
- Add `'rest_framework'` to `INSTALLED_APPS`
- Update `requirements.txt`

2. Create Serializers

Create `api/serializers.py` with:

- `CategorySerializer` — using `ModelSerializer`
- `ProductSerializer` — using `ModelSerializer`

3. Create ViewSets and Configure Router

a. ViewSets (`api/views.py`)

- `CategoryViewSet` extending `viewsets.ModelViewSet`
- `ProductViewSet` extending `viewsets.ModelViewSet`
- Add custom action for `/api/categories/<id>/products/`

b. Router (`api/urls.py`)

- Use `DefaultRouter` to register ViewSets
- Register `'categories'` and `'products'` endpoints

4. Register Models in Django Admin

- Register `Category` and `Product` in `api/admin.py`
- Create superuser and add **4 categories + 20 products**

5. Create Postman Collection

Create a collection named **"Online Shop API"** with two folders:

a. Categories folder

Request Name	Method	URL
Get All Categories	GET	<code>/api/categories/</code>
Get Category by ID	GET	<code>/api/categories/1/</code>
Get Products by Category	GET	<code>/api/categories/1/products/</code>

Create Category	POST	/api/categories/
Update Category	PUT	/api/categories/1/
Delete Category	DELETE	/api/categories/1/

b. Products folder

Request Name	Method	URL
Get All Products	GET	/api/products/
Get Product by ID	GET	/api/products/1/
Create Product	POST	/api/products/
Update Product	PUT	/api/products/1/
Delete Product	DELETE	/api/products/1/

c. Export Collection

- Export as `OnlineShopAPI.postman_collection.json`
- Include in your GitHub submission

Note: For POST/PUT requests, use **Body** → **raw** → **JSON** format.

Requirements

- `django-rest-framework` installed and configured
- `ModelSerializer` for Category and Product
- `ModelViewSet` with `DefaultRouter`
- All models registered in admin
- 4 categories, 20 products
- **Postman collection** with **11 requests** (6 Categories + 5 Products)
- **Exported JSON file** included in submission

Useful Links

1. DRF Official: [django-rest-framework.org](https://www.django-rest-framework.org)
2. DRF ViewSet Actions: [Extra Actions](#)
3. Postman Download: postman.com/downloads
4. Postman Collections: [Collections Overview](#)

GOOD LUCK! :)